Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Measurements of Earth's gravity field reveal how water, air, and the solid Earth redistribute mass. Ground-based gravimetry and gradiometry measure gravitational acceleration and its spatial gradients, respectively, while precise leveling combined with gravity observations determines potential differences.[1,2] Satellite missions have tracked large-scale changes in water storage and the climate system.[3,4] Monitoring time-varying potential differences between distant stations requires resolving weak signals and distinguishing contributions with similar temporal characteristics.[5] Tides provide a well-defined test case for this challenge. The tidal fields of the Moon and Sun act together with solid Earth deformation and the redistribution of ocean water. Atmospheric pressure tides also drive the oceans.[6] The solid Earth tidal potential can reach amplitudes of several $\mathrm{m}^2/\mathrm{s}^2$ at individual stations, and resolving smaller contributions requires higher measurement sensitivity.[7]

Optical clocks provide another means of measuring potential differences through their relative clock rates. Near the Earth's surface, a relative frequency change of 10^{-18} corresponds to an equivalent height of about 1 cm.[8] Experiments have resolved gravitational redshifts on laboratory scales and measured static potential differences between distant sites.[2,9,10,11] The fractional systematic uncertainty of optical clocks is now approaching 10^{-19}, and independent lattice-clock experiments have achieved instabilities of a few $10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is the averaging time.[12,13] Non-tidal mass changes in the atmosphere, oceans, and continental waters may be significant at the 10^{-18} fractional frequency level.[7] Advances in fiber-optic networks and free-space frequency transfer are extending clock comparisons to long distances.[14,15,16,17,18] These advances are driving the development of optical-clock networks for monitoring time-varying gravitational potentials.[5,8] However, resolving neighbouring tidal components requires sufficient sensitivity and temporal coverage, especially in the presence of observational gaps.

Here we detect tidal variations in the potential difference between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan. The clocks are separated by 692 km and connected by a 1,350-km fiber-optic link.[19] Using 545.9 hours of observations distributed over 353 days, we combine variations within observing segments while allowing each segment an independent frequency offset. We jointly estimate the responses to two neighbouring semidiurnal tidal groups, M_2+N_2 and S_2+K_2, with phases and relative amplitudes fixed within each group.

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